

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Using block diagram reduction technique find closed loop transfer function $C(s) / R(s)$	Level-5	6
B) Using Mason's gain formula obtain $C(s)/R(s)$ for the signal flow graph shown in fig	Level-5	6
C) Obtain Transfer function model for Mechanical System	Level-2	6
Q.2 Solve Any Two of the following.		12
A) Find k_p , k_v , k_a and steady state error for a system with open loop transfer function as $G(s)H(s) = \frac{15(s+4)(s+7)}{s(s+3)(s+6)(s+8)}$ Where the input is $r(t)=4+t+t^2$	Level-5	6
B) Utilize the Routh table to determine the number of roots of the following polynomials in the right half of the s-plane. Comment about the stability of the system	Level-3	6

$$(i) s^5 + 2s^4 + 3s^3 + 6s^2 + 10s + 15$$

$$(ii) s^5 + 6s^4 + 15s^3 + 30s^2 + 44s + 24$$

- C) Define Delay time, Rise time, Peak time, Peak overshoot, Settling time **Level-2** **6**

Q. 3 Solve Any Two of the following. **12**

- A) Sketch the root locus diagram of the following open loop transfer function **Level-4** **6**
 $G(s)H(s) = K/S(S+2)(S+5)$
- B) With suitable block diagrams and equations, explain the following types of controllers employed in control systems: **Level-2** **6**
 a) Proportional controller b) Proportional-plus-integral controller
 c) PID controller
- C) The unity feedback system is characterized by an open loop transfer function, $G(s) = K/S(S+10)$. **Level-3** **6**
 Determine the gain K, so that the system will have a damping ratio of 0.5. For this value of K, determine settling time, peak overshoot and time to peak overshoot for a unit step input

Q.4 Solve Any Two of the following. **12**

- A) Sketch the Nyquist stability plot for a feedback system with the following open-loop transfer function **Level-4** **6**

$$G(s)H(s) = \frac{1}{S(S^2 + S + 1)}$$

- B) What are the basic characteristics of feedback control system? Explain in details **Level-2** **6**
- C) For the following T.F draw the Bode plot and obtain Gain cross over frequency (wgc), Phase cross over frequency, Gain Margin and Phase Margin. **Level-4** **6**

$$G(s) = 20 / [s(1+3s)(1+4s)]$$

Q. 5 Solve Any Two of the following. **12**

- A) Test the controllability and observability of the system whose state space representation is given as **Level-4** **6**

$$\begin{bmatrix} \dot{X}_1 \\ \dot{X}_2 \\ \dot{X}_3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ -2 & -3 & 0 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} u; y = [1 \quad 0 \quad 0] \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}$$

- B) A discrete time is described by the difference equation **Level-5** **6**
 $y(k+2)+5y(k+1)+6y(k)=u(k)$, $y(0)=y(1)=0$; $T=1$ sec,
 Determine
 (i) State model in canonical form.
 (ii) State transition matrix
- C) Explain Lead, Lag and Lag-Lead compensators with a neat diagram **Level-2** **6**